

Tonsillar Microfold Cells: Structure, Function, Origin, Markers, and Single-Cell Transcriptomics

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Background: M cells in mucosal lymphoid tissues

M cells are specialized epithelial cells found in mucosal lymphoid tissues that move material from the lumen to immune cells below. Most of what is known comes from gut M cells, where they arise from epithelial precursors under strong control by RANKL-RANK signaling. (4 sources)

Microfold (M) cells are specialized epithelial cells in mucosa-associated lymphoid tissues (MALT). Their core job is to take up luminal microbes, particles, and other antigens and pass them to immune cells underneath, helping start mucosal immune responses. This role is described across MALT sites, including the gut and tonsils. ([Siddiqui et al., 2026](#)) In the best-studied setting, the gut, M cells are part of the follicle-associated epithelium (FAE) over gut-associated lymphoid tissue (GALT), where they make up about 10% of FAE cells and are distinct from neighboring absorptive epithelial cells. Their defining feature is strong phagocytic and transcytotic capacity for macromolecules, particulate antigens, and microorganisms, which makes them important for moving orally acquired pathogens and antigens to antigen-presenting cells for efficient immune responses. ([Walsh et al., 2014](#)) ([Mabbott et al., 2013](#))

Developmentally, the general model from intestinal M cells is that they come from epithelial stem or precursor cells rather than from hematopoietic cells. In the intestine, Lgr5+ epithelial stem cells in the FAE crypt region can give rise to M cells along with other epithelial lineages. ([Walsh et al., 2014](#)) ([Mabbott et al., 2013](#)) A notable principle is that local lymphoid tissue architecture helps instruct this fate: subepithelial stromal cells provide RANKL, epithelial precursors express RANK, and loss or blockade of RANKL largely eliminates M cells, while exogenous RANKL can restore or induce them. ([Walsh et al., 2014](#)) ([Knoop et al., 2009](#)) This gut-based framework is often used as the starting point for thinking about M cells in other organized mucosal lymphoid tissues, including the tonsil.

Evidence

(Siddiqui et al., 2026)

[Taste Papillae-associated Salivary Gland Ducts Contribute to Immune Surveillance](#)

"Mucosa-associated lymphoid tissue (MALT) such as those in the gut epithelium, tonsils and lachrymal and salivary ducts possess specialized cells called microfold (M) cells that transport luminal microbes and present them to underlying immune cells, which then initiate an appropriate immune response."

(Walsh et al., 2014)

Biology of the RANKL–RANK–OPG System in Immunity, Bone, and Beyond

"Microfold cells are specialized intestinal epithelial-derived cells that make up roughly 10% of the follicle-associated epithelial (FAE) cells covering the gut-associated lymphoid tissues (GALT). Microfold (M) cells exhibit some morphological distinctions compared to neighboring enterocytes, but are most important for their enhanced capacity for phagocytosis and transcytosis of macromolecules, particulate antigens, and micro-organisms residing in the gut lumen (Mabbott et al., 2013). M cells are essential for transfer of antigens from orally acquired pathogens, such as Salmonella Typhimurium, to antigen-presenting cells, and subsequently, for optimal immune responses (Mabbott et al., 2013)"

"Within the villous crypts of the FAE reside leucine-rich repeat-containing G-protein-coupled receptor-(Lgr5) expressing epithelial stem cells that are capable of differentiating into the various sub-lineage found in the gut, including enterocytes, goblet cells, enteroendocrine cells, tuft cells, Paneth cells, and M cells (Mabbott et al., 2013)"

"It has been shown that stromal cells in the subepithelial dome of the FAE provide RANKL to RANK-expressing stem cells to induce M cell differentiation, and that in RANKL-deficient mice, M cells fail to develop unless sufficient exogenous RANKL is provided (Knoop et al., 2011)(Knoop et al., 2009). This data demonstrated an absolute requirement for RANK signaling during M cell development. This work further suggested that RANK signaling on M cell precursors is triggered by membrane-bound RANKL on stromal cells, such that cell-cell contact and FAE micro-architecture may be critical to M cell differentiation (Knoop et al., 2011). RANKL has been shown to specifically induce the markers annexin A5 and myristoylated alanine-rich C-kinase substrate-like protein 1 (MARCKS) on M cell precursor stem cells, but their significance in this context remains unclear (9,73,74)."

(Mabbott et al., 2013)

Microfold (M) cells: important immunosurveillance posts in the intestinal epithelium

"The transcytosis of antigens across the gut epithelium by microfold cells (M cells) is important for the induction of efficient immune responses to some mucosal antigens in Peyer's patches. Recently, substantial progress has been made in our understanding of the factors that influence the development and function of M cells. This review highlights these important advances, with particular emphasis on: the host genes which control the functional maturation of M cells; how this knowledge has led to the rapid advance in our understanding of M-cell biology in the steady state and during aging; molecules expressed on M cells which appear to be used as "immunosurveillance" receptors to sample pathogenic microorganisms in the gut; how certain pathogens appear to exploit M cells to infect the host; and finally how this knowledge has been used to specifically target antigens to M cells to attempt to improve the efficacy of mucosal vaccines."

(Knoop et al., 2009)

RANKL is necessary and sufficient to initiate development of antigen-sampling M cells in the intestinal epithelium

"Microfold cells (M cells) are specialized epithelial cells situated over Peyer's patches (PP) and other organized mucosal lymphoid tissues that transport commensal bacteria and other particulate Ags into intraepithelial pockets accessed by APCs. The TNF superfamily member receptor activator of NF- κ B ligand (RANKL) is selectively expressed by subepithelial stromal cells in PP domes. We found that RANKL null mice have <2% of wild-type levels of PP M cells and markedly diminished uptake of 200 nm diameter fluorescent beads. Ab-mediated neutralization of RANKL in adult wild-type mice also eliminated most PP M cells. The M cell deficit in RANKL null mice was corrected by systemic administration of exogenous RANKL. Treatment with RANKL also induced the differentiation of villous M cells on all small intestinal villi with the capacity for avid uptake of Salmonella and Yersinia organisms and fluorescent beads. The RANK receptor for RANKL is expressed by epithelial cells throughout the small intestine. We conclude that availability of RANKL is the critical factor controlling the differentiation of M cells from RANK-expressing intestinal epithelial precursor cells."

Tonsillar location and tissue organization

In humans, M cells are part of the epithelium of Waldeyer's ring, especially the crypt and follicle-associated epithelium of the tonsils. Their position is important because tonsils lack afferent lymphatics, so crypts and M cells help bring inhaled or luminal antigens straight to the immune cells below. (6 sources)

In humans, the nasal-associated lymphoid tissue corresponds to Waldeyer's ring, which includes the nasopharyngeal, tubal, palatine, and lingual tonsils. This tissue is organized in a way that is broadly similar to Peyer's patches, with lymphoid follicles lying under follicle-associated epithelium that contains interspersed M cells. ([Brugman et al., 2015](#))

Within the tonsil, the epithelial surface is not smooth. Instead, the tonsillar epithelium invaginates deeply to form crypts, and beneath the epithelium lies a subepithelial compartment with multiple lymphoid follicles and distinct B- and T-cell areas. The reticular epithelium over these follicles contains not only M cells, but also lymphocytes, macrophages, dendritic cells, and some granulocytes, placing antigen-sampling epithelium in

direct contact with immune cells. ([Brugman et al., 2015](#)) ([Ganger et al., 2018](#))

Reports place tonsillar M cells in specialized folds of the crypt surface and in channels that open toward the lumen; some of these channels are directly exposed, while others are covered by flat M cells. This means that the crypt epithelium is a key portal where inhaled or luminal material can meet the mucosal immune system. ([Liu et al., 2024](#)) ([Hansen et al., 2026](#))

This location likely matters even more in the tonsil because, unlike lymph nodes, tonsils do not have an afferent lymphatic network. Instead, crypt architecture and M-cell-bearing epithelium help deliver antigens across the surface to underlying antigen-presenting cells and then into local immune reactions, including germinal center responses. ([Liu et al., 2024](#)) ([Kwon et al., 2026](#)) ([Hansen et al., 2026](#))

Tonsils also develop early as secondary lymphoid organs: B and T cells populate the area under the tonsillar epithelium by about 14 weeks of gestation, and primary follicles appear by 16 weeks, earlier than in other secondary lymphoid tissues. This early pairing of epithelium, crypts, and lymphoid follicles helps explain why the tonsil is such a structured antigen-sampling site. ([Brugman et al., 2015](#)) ([Mold et al., 2008](#))

Evidence

([Brugman et al., 2015](#))

[Mucosal Immune Development in Early Life: Setting the Stage](#)

"The NALT (named Waldeyer's ring in humans), consists of the nasopharyngeal tonsil, tubal tonsils, palatine tonsils and lingual tonsils (Perry and Whyte 1998). Its appearance is similar to Peyer's patches; follicles underneath follicle-associated epithelium containing interspersed microfold cells that can sample antigens (Breel et al. 1988a, b). Tonsils are secondary lymphoid organs. The tonsillar subepithelial space is formed by several lymphoid follicles containing B and T cell areas. Tonsils are not encapsulated like the spleen, but are lined by tonsillar epithelium that invaginates forming crypts (Perry and Whyte 1998). From the 14th week of gestation, B and T cells populate the area under the tonsillar epithelium and primary follicles develop from 16 weeks of gestation (earlier than any other secondary lymphoid tissue)."

([Ganger et al., 2018](#))

[Tailoring Formulations for Intranasal Nose-to-Brain Delivery: A Review on Architecture, Physico-Chemical Characteristics and Mucociliary Clearance of the Nasal Olfactory Mucosa](#)

"The reticular epithelium overlying the tonsils with its lymphoid follicles contains Microfold cells, lymphocytes, macrophages, dendritic cells and some granulocytes."

([Liu et al., 2024](#))

[Effects of tonsillectomy and adenoidectomy on the immune system](#)

"Some superficial channels open directly on the surface, whereas some channels are covered with flat microfold cells (M cells). As the tonsils have no afferent lymphatic network, the presence of crypts and M cells is particularly important for the regulation of immune functions."

([Hansen et al., 2026](#))

[Nasal humoral immunity: the first line of protection against respiratory pathogens](#)

"The outer epithelial cell layer of tonsils has epithelial cells known as microfold (M) cells, which are present in specialized folds in the surface of the tonsil crypts. (Fujimura, 2000) Once antigen enters into tonsils via M cells, innate and adaptive immune cells can become activated and drive the formation of germinal center (GC) responses."

([Kwon et al., 2026](#))

[Harnessing mucosal immunity for protective vaccines](#)

"The epithelial layer covering the NALT and tonsils contains microfold cells that translocate inhaled antigens to the underlying antigen-presenting cells (APCs)."

([Mold et al., 2008](#))

[Maternal Alloantigens Promote the Development of Tolerogenic Fetal Regulatory T Cells in Utero](#)

"As the immune system develops, T cells are selected or regulated to become tolerant of self antigens and reactive against foreign

antigens. In mice, the induction of such tolerance is thought to be attributable to the deletion of self-reactive cells. Here, we show that the human fetal immune system takes advantage of an additional mechanism: the generation of regulatory T cells (Tregs) that suppress fetal immune responses. We find that substantial numbers of maternal cells cross the placenta to reside in fetal lymph nodes, inducing the development of CD4+CD25highFoxP3+ Tregs that suppress fetal antimaternal immunity and persist at least until early adulthood. These findings reveal a form of antigen-specific tolerance in humans, induced in utero and probably active in regulating immune responses after birth."

Structure and immune function of tonsillar M cells

Tonsillar M cells are flattened epithelial cells built for antigen uptake rather than barrier coating. They sample inhaled or luminal material at the tonsil surface and move it to immune cells just below, helping start local IgA and other mucosal immune responses. (7 sources)

Tonsillar M cells are thought to share the core structural design seen in M cells across mucosal lymphoid tissues: they have a reduced or absent brush border at the apical surface, giving the characteristic "microfold" appearance rather than the dense microvilli seen on ordinary epithelial cells. They also tend to lack the thick apical mucus or glycocalyx barrier that covers many neighboring epithelial cells, which helps explain why particles and microbes can contact them more easily than conventional epithelial cells. On the basolateral side, M cells form close contact zones or "pockets" with immune cells, including dendritic cells, macrophages, and lymphocytes, creating a short path from surface uptake to immune presentation. Although much of this structural detail was worked out in intestinal M cells, the same general functional logic is used to interpret M cells in tonsillar follicle-associated epithelium. ([Menard et al., 2022](#)) ([Dillon et al., 2019](#)) ([Frey et al., 1996](#))

Functionally, the main job of tonsillar M cells is rapid uptake and transepithelial transport of antigens, microorganisms, and particles from the crypt or luminal surface to the lymphoid tissue underneath. This makes them a front-line sampling system in Waldeyer's ring, where they can help initiate effective mucosal immune responses rather than simply act as a passive barrier. Work across respiratory and nasopharyngeal lymphoid tissues supports the idea that these cells are key for sampling inhaled antigens and for promoting antigen-specific responses, especially IgA-linked mucosal immunity. ([Kojima et al., 2012](#)) ([Abdel-Razek et al., 2022](#)) ([Kiyono et al., 2004](#))

A useful way to think about tonsillar M cells is that their structure is tuned for controlled barrier breach. Their sparse apical surface makes microbial or particle binding easier, while their basolateral association with immune cells allows fast delivery of sampled material into local immune circuits. This same property also makes M cells potential entry points for pathogens, which is why they are often discussed both as protective antigen-sampling cells and as portals that microbes may exploit. ([Menard et al., 2022](#)) ([Dillon et al., 2019](#)) ([Kumar, 2025](#))

In the tonsil specifically, interest has also focused on the glycoconjugates present on the apical membrane of follicle-associated epithelium and M cells. These surface molecules may serve as initial binding sites for lectins and microbial surface proteins, which helps explain both selective antigen uptake and the idea of targeting tonsillar M cells for mucosal vaccine delivery. This has made tonsillar and nasopharyngeal M cells attractive as experimental targets for nasal or oral vaccine strategies. ([Kumar, 2025](#)) ([Kojima et al., 2012](#))

Evidence

([Menard et al., 2022](#))

[Cross-Talk Between the Intestinal Epithelium and Salmonella Typhimurium](#)

"M cells represent intestinal cells found in a small proportion of follicular-associated epithelium covering mucosal lymphoid tissues, such as large aggregates of B lymphocyte follicles called colonic patches, caecal patches or Peyer's patches (PPs) in the small intestine. Their cell structure differs from the other intestinal cells. Indeed, microvilli characteristics of enterocytes are missing at the apical surface of M cells, resulting in a « microfold » morphology, hence the name of microfold or M cells (Dillon et al., 2019). In addition, the mucus layer that covers other epithelial cells is absent from the apical surface of M cells (Frey et al., 1996). At the basolateral side, the M cells have a pocket containing dendritic cells, macrophages, and T and B cells allowing the initiation of an

appropriate mucosal immune response (Neutra et al., 2001). A direct interaction of the B cells with M cells is an essential step required for the acquisition of M cell function (Hsieh et al., 2010). These features promote the role of the M cells in the capture of luminal particles and antigens, the transport through the epithelium and the release to submucosal tissues (Tam et al., 2008)"

"M celllike cells have also been described at the tips of intestinal villi and are thus called "villous M cells" (Dillon et al., 2019). They show morphological similarity to PP M cells with the characteristic absence of apical microvilli and they can be stained by the lectin *Ulex Europaeus* Agglutinin-1 which binds to fucosylated molecules. Their origin would come from a trans-differentiation rather than from crypt stem cells like other epithelial cells. The cellular signals or cytokines that are responsible for this trans-differentiation are still undefined."

(Dillon et al., 2019)

M Cells: Intelligent Engineering of Mucosal Immune Surveillance

"M cells are specialized intestinal epithelial cells that provide the main machinery for sampling luminal microbes for mucosal immune surveillance. M cells are usually found in the epithelium overlying organized mucosal lymphoid tissues, but studies have identified multiple distinct lineages of M cells that are produced under different conditions, including intestinal inflammation. Among these lineages there is a common morphology that helps explain the efficiency of M cells in capturing luminal bacteria and viruses; in addition, M cells recruit novel cellular mechanisms to transport the particles across the mucosal barrier into the lamina propria, a process known as transcytosis. These specializations used by M cells point to a novel engineering of cellular machinery to selectively capture and transport microbial particles of interest. Because of the ability of M cells to effectively violate the mucosal barrier, the circumstances of M cell induction have important consequences. Normal immune surveillance insures that transcytosed bacteria are captured by underlying myeloid/dendritic cells; in contrast, inflammation can induce development of new M cells not accompanied by organized lymphoid tissues, resulting in bacterial transcytosis with the potential to amplify inflammatory disease. In this review, we will discuss our own perspectives on the life history of M cells and also raise a few questions regarding unique aspects of their biology among epithelia."

(Frey et al., 1996)

Role of the glycocalyx in regulating access of microparticles to apical plasma membranes of intestinal epithelial cells: implications for microbial attachment and oral vaccine targeting

"Trans epithelial transport of antigens and pathogens across the epithelial barrier by M cells may be a prerequisite for induction of mucosal immunity in the intestine. Efficient transport of antigens and pathogens requires adherence to M cell apical surfaces. Coupling of antigen-containing particles to the pentameric binding subunit of cholera toxin (CTB) has been proposed as a means for increasing antigen uptake because the CTB receptor, ganglioside GM1, is a glycolipid present in apical membranes of all intestinal epithelial cells. To test the accessibility of enterocyte and M cell membrane glycolipids to ligands in the size ranges of viruses, bacteria, and particulate mucosal vaccines, we analyzed binding of CTB probes of different sizes to rabbit Peyer's patch epithelium. Soluble CTB-fluorescein isothiocyanate (diameter 6.4 nm) bound to apical membranes of all epithelial cells. CTB coupled to 14 nm colloidal gold (final diameter, 28.8 nm) failed to adhere to enterocytes but did adhere to M cells. CTB-coated, fluorescent microparticles (final diameter, 1.13 microns) failed to adhere to enterocytes or M cells in vivo or to well-differentiated Caco-2 intestinal epithelial cells in vitro. However, these particles bound specifically to GM1 on BALB/c 3T3 fibroblasts in vitro and to undifferentiated Caco-2 cells that lacked brush borders and glycocalyx. Measurements of glycocalyx thickness by electron microscopy suggested that a relatively thin (20 nm) glycocalyx was sufficient to prevent access of 1-micron microparticles to glycolipid receptors. Thus, the barrier function of the intestinal epithelial cell glycocalyx may be important in limiting microbial adherence to membrane glycolipids, and in CTB-mediated targeting of vaccines to M cells and the mucosal immune system."

(Kojima et al., 2012)

Regulation of Tight Junctions in Upper Airway Epithelium

"M cells (membranous or microfold cells) are known as specialized epithelial cells of the follicle-associated epithelium (FAE), and the role seems to be the rapid uptake of particular antigens and microorganisms to the immune cells of the lymphoid follicle to induce an effective immune response (Mach et al., 2005). The structure and functional peculiarities of M cells are observed in different species and at different sites of the lymphoid tissue along the digestive and respiratory tracts (Gebert, 1997). The respiratory M cells act as a nasopharynx-associated lymphoid tissue in the upper respiratory tract (Kim et al., 2011)"

"Several markers such as lectin histochemistry and immunoreactivity to vimentin, cytokeratins, and annexin-V have been proposed to identify M cells in the mouse, rat, hamster, rabbit, and pig (Mach et al., 2005)(Gebert, 1997)(Giannasca et al., 1997)(Kucharzik et al., 1998)(Verbrugghe et al., 2006). In the nasopharyngeal tonsil of the horse, lectin GS-1 B4 (*Griffonia simplicifolia* 1 isolectin B4) has been used as a marker for M cells (Kumar et al., 2001). It was reported that cytokeratin20 (Ck20) could serve as an M cell marker for rabbit palatine tonsils (Carapelli et al., 2004). In humans, clusterin is expressed in M cells and follicular dendritic cells at inductive sites of human mucosa-associated lymphoid tissue (Verbrugghe et al., 2008). Class II beta-tubulin is a specific histochemical marker for human tonsillar M cells and follicular dendritic cells [55]. However, universal markers for human M cells have not yet been established."

(Abdel-Razek et al., 2022)

Surfactant proteins and innate immunity of otitis media

"Inductive sites are organized in mucosal-associated lymphoid tissue (MALT), which is known more specifically in the nasal area as nasal associated lymphoid tissue (NALT). NALT consists of the adenoids and palatine tonsils which form Waldeyer's ring in humans. (Neutra et al., 2006)(Sabirov et al., 2008)"

"Recently, respiratory microfold (M) cells were identified in the murine airways. (Macarthur et al., 2011) Respiratory M cells were capable of sampling inhaled bacterial antigens to initiate an Ag-specific immune response. It is widely accepted that NALT M cells are key players in the uptake of nasally delivered antigens for the subsequent induction of antigen-specific IgA immune responses. (Kiyono et al., 2004)(Kim et al., 2011)"

(Kiyono et al., 2004)

NALT- versus PEYER'S-patch-mediated mucosal immunity

"Recent studies indicate that the mechanism of nasopharynx-associated lymphoid tissue (NALT) organogenesis is different from that of other lymphoid tissues. NALT has an important role in the induction of mucosal immune responses, including the generation of T helper 1 and T helper 2 cells, and IgA-committed B cells. Moreover, intranasal immunization can lead to the induction of antigen-specific protective immunity in both the mucosal and systemic immune compartments. Therefore, a greater understanding of the differences between NALT and other organized lymphoid tissues, such as Peyer's patches, should facilitate the development of nasal vaccines."

(Kumar, 2025)

Novel Insights into Lectin Binding Patterns in the Nasopharyngeal Tonsil of Buffaloes (Bubalus bubalis)

"Within the FAE, specialized microfold (M) cells have been identified in the nasopharyngeal tonsils of horses, pigs, and buffaloes (Kumar et al., 2010)(Ranjit et al., 2015)[4]. M-cells play a crucial role in generating mucosal immune responses by transcytosis antigens from the free epithelial surface of the tonsil. M-cells can shed light on early pathogenesis by probing the interactions between microorganisms and the unique glycoconjugate receptors in the future. The expression of glycoconjugates specific to the FAE and M-cell surfaces may serve as an experimental model for developing mucosal vaccines targeting M-cells to prevent infections (Giannasca et al., 1994). The apical membrane conjugates of the FAE, particularly M-cells, may serve as initial binding sites for lectins and microbial surface proteins (Clark et al., 1993). Furthermore, the expression of glycoconjugates differs significantly in the FAE and lymphoid tissue between mice and humans (Sharma et al., 1996)."

Developmental origin and differentiation

Direct evidence for how human tonsillar M cells develop is still limited. The best current airway lymphoid tissue data suggest that human upper-airway M cells can arise from epithelial progenitors related to club cells, rather than from immune-cell lineages, and may follow a program that is not identical to intestinal M cells. (2 sources)

For tonsil specifically, the literature still treats M-cell development more as an informed inference than a fully mapped pathway. Recent review-level discussion notes that work on tubal-associated lymphoid tissue (TALT) M cells led investigators to speculate that a comparable antigen-sampling M-cell phenotype also exists in the human tonsil, but this does not by itself define its lineage path in tonsillar tissue. ([Castillo et al., 2024](#))

The strongest human upper-airway developmental evidence comes from adenoid, a related Waldeyer's ring tissue. Using single-nucleus RNA-seq and trajectory analysis, Alvarez-Arguedas et al. reported that human airway M cells develop from progenitor club cells and that they show a gene signature distinct from intestinal M cells. This is important because it supports an epithelial origin for airway M cells, but suggests that the differentiation route in the nasopharyngeal/tonsillar region may differ from the classic gut model rather than simply copying it. ([Alvarez-Arguedas et al., 2024](#))

Taken together with the gut framework summarized above, the safest current view is that tonsillar M cells are likely epithelial derivatives shaped by the local lymphoepithelial niche, but the exact precursor state, intermediate steps, and inductive signals in human tonsil remain incompletely defined. In other words, airway-associated M-cell biology appears related to, but not the same as, intestinal M-cell differentiation. ([Alvarez-Arguedas et al., 2024](#)) (Model-Generated)

Evidence

(Castillo et al., 2024)

Deciphering the M-cell niche: insights from mouse models on how microfold cells “know” where they are needed

"Oya et al., who characterized the TALT M cells, speculated on the existence of an M-cell phenotype for sampling in the human tonsil"

(Alvarez-Arguedas et al., 2024)

Single cell transcriptional analysis of human adenoids identifies molecular features of airway microfold cells

"MALT contains a variety of specialized epithelial cells, including a rare cell type called a microfold cell (M cell) that functions to transport apical antigens to basolateral antigen-presenting cells, a crucial step in the initiation of mucosal immunity. M cells have been extensively characterized in the gastrointestinal (GI) tract in murine and human models. However, the precise development and functions of human airway M cells is unknown. Here, using single-nucleus RNA sequencing (snRNA-seq), we generated an atlas of cells from the human adenoid and identified 16 unique cell types representing basal, club, hillock, and hematopoietic lineages, defined their developmental trajectories, and determined cell-cell relationships. Using trajectory analysis, we found that human airway M cells develop from progenitor club cells and express a gene signature distinct from intestinal M cells."

Markers used to identify tonsillar or airway M cells

No single marker fully defines tonsillar or airway M cells, so studies usually use marker combinations. In human tonsil and adenoid datasets, the most useful current markers include epithelial keratins such as CK8/18 and transcriptional markers such as **SPIB** and **MARCKSL1**, with additional mature and immature gene sets from airway single-cell work. (3 sources)

- **CK8/18 (cytokeratin 8/18)**: immunohistochemical work has used CK8/18 as a practical marker to detect M cells in mucosal lymphoid epithelium, including in studies cited for tonsillar M cells. Perry et al. specifically note that CK8/18 is useful for detecting microfold cells and cite earlier M-cell studies for this use. ([Perry et al., 2025](#))
- **SPIB**: this is one of the clearest canonical transcriptional markers carried over from the broader M-cell field into human upper-airway and tonsillar analysis. In the adenoid/tonsil comparison, Alvarez-Arguedas et al. found SPIB expression in a subset of tonsillar epithelial “crypt cells,” supporting the idea that at least a fraction of crypt epithelium has an M-cell-like program. ([Alvarez-Arguedas et al., 2025](#))
- **MARCKSL1**: alongside SPIB, MARCKSL1 was detected as a canonical M-cell marker in a subset of human tonsillar crypt epithelial cells, making it another useful transcript-level marker when looking for M-cell-like states in Waldeyer’s ring tissues. ([Alvarez-Arguedas et al., 2025](#))
- **“Mature M-cell” DEG sets from airway single-cell data**: in adenoid snRNA-seq, mature M cells were defined by a broader differentially expressed gene program rather than by one marker alone. When that program was compared with published tonsil epithelial data, most mature adenoid M-cell DEGs were found within tonsillar “crypt cells,” although some genes seen in adenoid mature M cells were not detected in tonsil, including **CEMIP**, **XKR4**, and **TMEM178B**. This suggests that marker panels may need to be tissue-specific and that human tonsillar M cells may only partially match airway adenoid M cells at the transcript level. ([Alvarez-Arguedas et al., 2025](#)) ([Massoni-Badosa et al., 2022](#))
- **“Immature M-cell” markers from airway single-cell data**: Alvarez-Arguedas et al. also defined immature M-cell states in adenoid. In the human tonsil dataset they examined, only **DOCK8**, **PLGC2**, and **TNFAIP8** from the immature M-cell DEG list were significantly expressed, and these were seen across many epithelial populations rather than only in a discrete M-cell cluster. That means these genes may be useful as part of a developmental panel, but are less specific than canonical mature markers like SPIB. ([Alvarez-Arguedas et al., 2025](#)) ([Massoni-Badosa et al., 2022](#))
- **Practical take-home point**: for tonsil or airway tissue, the safest approach is to use combinations of markers plus location in the crypt/follicle-associated epithelium, rather than rely on any one stain or transcript. Right now, a reasonable working panel from the quoted literature is **CK8/18** for histology together with **SPIB**, **MARCKSL1**, and a broader mature/immature M-cell gene signature for transcriptomic studies. ([Perry et al., 2025](#)) ([Alvarez-Arguedas et al., 2025](#))

Paper	M Cell Marker Identification Methodology	List Of Identified Markers	M Cell Function Analysis	Marker Specificity	Developmental Origin Of M Cells	Location Of M Cells
Single cell transcriptional analysis of human adenoids identifies molecular features of airway microfold cells	Single-nucleus RNA sequencing (snRNA-seq) and trajectory analysis.	Gene signature distinct from intestinal M cells.	Transport apical antigens to basolateral antigen-presenting cells.	Unique gene signature distinct from intestinal M cells.	M cells develop from progenitor club cells.	Human airway M cells identified in the adenoid.

Evidence

(Perry et al., 2025)

[Modulation of cytokeratin and cytokine/chemokine expression following influenza virus infection of differentiated human tonsillar epithelial cells](#)

"Additionally, CK8/18 serves as a useful marker for detecting microfold cells, which play a role in antigen uptake and presentation by the immune system (Gebert et al., 1994)(Hondo et al., 2011)."

(Alvarez-Arguedas et al., 2025)

[Single cell transcriptional analysis of human adenoids identifies molecular features of airway microfold cells](#)

"MALT contains a variety of specialized epithelial cells, including a rare cell type called a microfold cell (M cell) that functions to transport apical antigens to basolateral antigen-presenting cells, a crucial step in the initiation of mucosal immunity. M cells have been extensively characterized in the gastrointestinal (GI) tract in murine and human models. However, the precise development and functions of human airway M cells are unknown"

". While M cells have previously been described in the human palatine tonsil through ultrastructural and immunohistochemical studies, [122][123][124][125] a transcriptionally defined M cell cluster was not specifically annotated in the recent single-cell atlas of the tonsil epithelium. 19 Nevertheless, expression of canonical M cell markers such as SPIB and MARCKSL1 was detected in a subset of epithelial cells termed 'crypt cells'. When we interrogated all human tonsillar epithelial cells, we found that the majority of mature M cell DEGs from our analysis in the adenoid were expressed by 'crypt cells', although some adenoid M cell DEGs were absent such as CEMIP, XKR4 and TMEM178B (Fig. 9A).

Notably, the percentage of crypt cells that expressed mature M cell DEGs was very low (typically below 20%), while other tonsillar populations more robustly expressed some M cell markers. In the case of the immature M cell DEGs, we found that only DOCK8, PLGC2, and TNFAIP8 were significantly expressed in the dataset by almost all the human epithelial cell populations in the tonsil"

". Here, using single-nucleus RNA sequencing (snRNA-seq), we generated an atlas of cells from the human adenoid and identified 26 unique cell types representing basal, club, hillock, and hematopoietic lineages, defined their developmental trajectories, and determined cell-cell relationships. Using trajectory analysis, we found that human airway M cells develop from progenitor club cells and express a gene signature distinct from intestinal M cells"

". In this work, using single cell nuclear transcriptomics, we provide a detailed annotation of both immune and non-immune cells within the human adenoid. Furthermore, by analyzing the data using bioinformatic tools, we identified several unexpected cell types and states in the human adenoid, including an unknown epithelial cell type with a robust interferon signature and immature and mature M cells. We also used trajectory analysis to identify the stepwise maturation of cells within the adenoid."

(Massoni-Badosa et al., 2022)

[An atlas of cells in the human tonsil](#)

"Palatine tonsils are secondary lymphoid organs representing the first line of immunological defense against inhaled or ingested pathogens. Here, we present a comprehensive census of cell types forming the human tonsil by applying single-cell transcriptome, epigenome, proteome and adaptive immune repertoire sequencing as well as spatial transcriptomics, resulting in an atlas of >357,000 cells. We provide a glossary of 121 annotated cell types and states, and disentangle gene regulatory mechanisms that drive cells through specialized lineage trajectories. Exemplarily, we stratify multiple tonsil-resident myeloid slancye subtypes, establish a distant BCL6 superenhancer as locally active in both follicle-associated T and B cells, and describe SIX5 as a potentially novel transcriptional regulator of plasma cell maturation. Further, our atlas is a reference map to understand alterations observed in disease. Here, we discover immune-phenotype plasticity in tumoral cells and microenvironment shifts of mantle cell lymphomas (MCL). To facilitate such reference-based analysis, we develop HCATonsilData and SLOcator, a computational framework that provides programmatic and modular access to our dataset; and allows the straightforward annotation of future single-cell profiles from secondary lymphoid organs."

Single-cell transcriptomics evidence in tonsil/adenoid

Single-cell data from Waldeyer's ring tissues give mixed support for a distinct tonsillar M-cell population. Adenoid single-cell/nucleus work supports an airway M-cell program, but published palatine tonsil transcriptomic analyses have not consistently found a clean, separate M-cell cluster. (3 sources)

The current single-cell picture is therefore uneven across related tissues. In adenoid, recent single-cell and single-nucleus analyses support the presence of an airway M-cell-like epithelial program and were strong enough to define mature and immature M-cell gene sets, which is why those markers are now often carried into tonsil studies as a reference framework. By contrast, in a palatine tonsil transcriptomic study, Martin et al. found that markers previously used for adenoid and tonsil M cells, including class II β -tubulin and clusterin, were broadly expressed across epithelial cells, and that none of the epithelial clusters showed a specific M-cell signature. They therefore argued that antigen entry in the tonsillar crypt barrier may rely less on a clearly separable transcriptomic M-cell population and more on passive passage through the lymphoreticular epithelium or active transport by abundant myeloid cells in that epithelial barrier. ([Martin et al., 2023](#))

This negative or ambiguous tonsil transcriptomic result does not mean the tissue lacks epithelial specialization. Rather, it suggests that if tonsillar M cells exist as a discrete state, they may be rare, short-lived, transcriptionally blended with other crypt epithelial cells, or better defined by location and function than by a single clean cluster in standard scRNA-seq datasets. That idea fits with spatial single-cell protein mapping of human tonsil tissue by Pham et al., who identified about 67,000 cells and partitioned them into 10 clusters, with one epithelial-dominant cluster mapping mainly to the epithelium and other clusters localizing to germinal centers, lymph nodules, and connective tissue. This study gives useful spatial single-cell context for tonsil organization, but it did not resolve a dedicated M-cell cluster, reinforcing the view that current single-cell evidence for palatine tonsil M cells is still indirect compared with adenoid. ([Pham et al., 2021](#)) ([Amir et al., 2013](#))

Taken together, the safest reading of the single-cell literature is that adenoid provides the stronger transcriptomic evidence for an upper-airway M-cell program, while palatine tonsil datasets so far show crypt epithelium with partial or distributed M-cell-like features rather than an unambiguous, stand-alone M-cell cluster. For practical work, that means tonsillar M-cell identification still benefits from combining single-cell signatures with tissue location, epithelial context, and orthogonal staining rather than expecting one definitive transcriptomic cluster in every dataset. ([Martin et al., 2023](#))

Evidence

([Martin et al., 2023](#))

[PI16+ reticular cells in human palatine tonsils govern T cell activity in distinct subepithelial niches](#)

"Indeed, previously described (M) cell markers of adenoids (Ogasawara et al., 2011) and palatine tonsils such as class II β -tubulin 37 and clusterin (Verbrugghe et al., 2008) are broadly expressed across epithelial cells in our transcriptomic analysis, whereas none of the cell clusters revealed a specific M cell signature. Thus, antigenic material present in the oropharynx is most likely either passively pressed through the membranous lymphoreticular crypt epithelium as a liquid condensate or actively transported further

into the lymphoid tissue by myeloid cells that are abundant in the specialized epithelial barrier."

(Pham et al., 2021)

Ultrasensitive and Multiplexed Protein Imaging with Cleavable Fluorescent Tyramide and Antibody Stripping

"The generated single-cell in situ protein expression profiles also allow us to study cell heterogeneity and the spatial distribution of the various cell types in human tonsil tissues. To achieve that, we calculated the expression levels of the 20 examined proteins in each of ~67,000 cells identified in the tissue. Based on their unique protein expression patterns (Figure 5A and Supplementary Figure S1), those individual cells were partitioned into 10 different cell clusters (Figure 5B) using the software viSNE (Amir et al., 2013). We then mapped these 10 cell clusters back to their natural tissue locations (Figure 5C and Supplementary Figure S2) and observed that the varied subregions of tonsil tissue are composed of cells from distinct clusters. For instance, cluster 7 is the major cell type in epithelium. The germinal centers mainly consist of clusters 8 and 10, while the lymph nodules are dominated by clusters 5, 6 and 9. Clusters 1 and 4 only appear in connective tissues. These results indicate that our approach enables the study of cell-type classification and their spatial distribution in FFPE tissues."

(Amir et al., 2013)

viSNE enables visualization of high dimensional single-cell data and reveals phenotypic heterogeneity of leukemia

"New high-dimensional, single-cell technologies offer unprecedented resolution in the analysis of heterogeneous tissues. However, because these technologies can measure dozens of parameters simultaneously in individual cells, data interpretation can be challenging. Here we present viSNE, a tool that allows one to map high-dimensional cytometry data onto two dimensions, yet conserve the high-dimensional structure of the data. viSNE plots individual cells in a visual similar to a scatter plot, while using all pairwise distances in high dimension to determine each cell's location in the plot. We integrated mass cytometry with viSNE to map healthy and cancerous bone marrow samples. Healthy bone marrow automatically maps into a consistent shape, whereas leukemia samples map into malformed shapes that are distinct from healthy bone marrow and from each other. We also use viSNE and mass cytometry to compare leukemia diagnosis and relapse samples, and to identify a rare leukemia population reminiscent of minimal residual disease. viSNE can be applied to any multi-dimensional single-cell technology."